

Answer Key

26FL-MG628-158W - Data Analysis for Managerial Decision- Making

Week 3 Case study Assignments

Assignment Objective:

- Analyzing Unstructured vs. Semi-Structured Data

Note: MPA students can choose the Business or the Healthcare assignment depending on your area of focus.

Week 3 Case Study (Business Majors)

TPay Customer Feedback & Transaction Analysis Case Study

TPay, a fintech startup in Ethiopia, has launched a pilot mobile wallet. To improve, they need to look at two different types of information:

1. **What people say** (Customer Feedback - Unstructured).
2. **What the system does** (Transaction Logs - Semi-Structured).

Dataset

Dataset A: Unstructured Data (Customer Reviews)

Here are 10 customer reviews with varied feedback:

Notice: This data is "messy" because it is human language without a fixed format.

1. "The app is fast and simple, but sometimes it crashes when I try to send money."
2. "Customer service takes too long to respond. I waited 3 days for help with a failed transaction."
3. "Very useful for paying my electricity bill, but I wish it supported more merchants."
4. "I don't trust it yet. I'm worried about my data security and privacy."
5. "Love it! Much faster than going to the bank. Saved me so much time."
6. "The bill payment feature is excellent, but the app freezes whenever I try to check my transaction history."
7. "I sent money to the wrong number and customer support helped me reverse it quickly. Great service!"
8. "App interface is confusing. I accidentally paid twice for the same bill because the button didn't respond."
9. "Security features like PIN and fingerprint make me feel safe, but I wish there was two-factor authentication."
10. "Merchant payments work perfectly, but money transfers to other users fail about 30% of the time."

Dataset B: Semi-Structured Data (Transaction Logs - JSON Format)

```
[
  { "transaction_id": "TXN001", "user_id": "U123", "type": "money_transfer", "amount": 500, "status": "success", "timestamp": "2024-01-15T09:23:45Z" },
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Dataset B: Semi-Structured Data (Table Format):

Notice: This data is organized by "tags" or "keys," making it easy for computers to read.

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Week 3 Case study (Health Majors)

XYZ Healthcare Systems: Data-Driven Operations Audit

Overview:

- This case study assignment combines the **Efficiency Audit** (Supply Chain) with the **Mobile Pay Initiative** (Revenue Cycle) into one comprehensive Case Study for Healthcare Administration.
- As a Healthcare Administrator, you are responsible for the "Business of Care." This assignment requires you to look at two different operational challenges using both **Quantitative (Numbers)** and **Qualitative (Stories)** data.

The Dataset

Part 1: The Supplier Efficiency Audit (Supply Chain)

Objective: Analyze the "Transaction-to-Finalization" gap to optimize cash flow and vendor reliability.

The Dataset

Transaction ID	Supplier Name	Category	Trans. Date	Final. Date	Cycle Time (Days)
TXN-401	MedTech Solutions	Surgical Kits	Oct 01	Oct 05	4
TXN-402	Apex Pharma	Generic Drugs	Oct 02	Oct 18	?

TXN-403	Global PPE	Protective Gear	Oct 05	Oct 08	3
TXN-404	MedTech Solutions	Surgical Kits	Oct 10	Oct 14	4
TXN-405	Apex Pharma	Specialty Drugs	Oct 12	Nov 02	?
TXN-406	BioLink Lab	Lab Reagents	Oct 15	Oct 30	15
TXN-407	Global PPE	Protective Gear	Oct 20	Oct 22	2

Question 1

25 points

Business Majors

Dataset A: Unstructured Data (Customer Reviews)

Here are 10 customer reviews with varied feedback:

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- "Very useful for paying my electricity bill, but I wish it supported more merchants."
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- 7. "I sent money to the wrong number and customer support helped me reverse it quickly. Great service!"
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- 10. "Merchant payments work perfectly, but money transfers to other users fail about 30% of the time."

Question 1: Thematic Analysis.

- (1a) Using Dataset A: Unstructured Data (Customer Reviews), group the customer reviews into three categories: **Technical Issues**, **Customer Service**, and **Trust/Security**.
- (1b) **Sentiment Check:** Label each of the 10 reviews as either **Positive**, **Negative**, or **Mixed**.

Healthcare Majors

Part 1: The Supplier Efficiency Audit (Supply Chain)

The Supplier Efficiency Audit (Supply Chain) Dataset

Transaction ID	Supplier Name	Category	Trans. Date	Final. Date	Cycle Time (Days)
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TXN-406	BioLink Lab	Lab Reagents	Oct 15	Oct 30	15
TXN-407	Global PPE	Protective Gear	Oct 20	Oct 22	2

Question 1: Thematic Analysis.

In the table above, locate the two rows with transaction IDs TXN-402 and TXN-405 associated with Apex Pharma.

(1a) Calculate: Calculate the cycle times missing in the table rows for TXN-402 and TXN-405).

(1b) Label each of the four suppliers (MedTech Solutions, Apex Pharma, Global PPE, and BioLink Lab) with one of the following performance ratings based on their cycle time efficiency

- 1. High-Efficiency Suppliers** (Cycle time ≤ 4 days)
- 2. Moderate-Efficiency Suppliers** (Cycle time 5-10 days)
- 3. Low-Efficiency Suppliers** (Cycle time > 10 days)

(1c:) Sentiment Check : Label each of the three suppliers (MedTech Solutions, Apex Pharma, Global PPE, and BioLink Lab) with a reliability rating based on their transaction patterns:

- **Reliable** (Consistent, short cycle times; ≤ 5 days average)
- **Unreliable** (Long or highly variable cycle times; > 10 days or high variance)
- **Inconsistent** (Mixed performance; some fast, some slow transactions)

Question 2

25 points

Log Analysis: (Business Majors)

Log Analysis:

Dataset B: Semi-Structured Data (System JSON Logs)

This data is organized by "tags" or "keys," making it easy for computers to read.

Here's your data converted to CSV format:

JSON

```
transaction_id,user_id,type,amount,status
TXN001,U123,money_transfer,500,success
TXN002,U456,bill_payment,1200,failed
TXN003,U789,merchant_payment,850,success
```

Look at Dataset B: Semi-Structured Data (System JSON Logs)

- **(3a)** What is the Success Rate? (Number of success / Total transactions)
- **(3b)** Which specific service (e.g., electricity) had a failure?

Log Analysis: (Healthcare Majors)

The Technical Log (Semi-Structured JSON Data)

JSON

```
{ "transaction_id": "MOB-8821",  
  "status_code": "PENDING",  
  "payment_gateway": "Stripe",  
  "auth_status": "SUCCESS",  
  "processing_details": {  
    "gateway_confirmation": "Confirmed",  
    "internal_sync_status": "FAILED",  
    "error_log": "HIS_Handshake_Timeout"  
  }  
}
```

- **(3a) Evidence:** Patient B says the system "doesn't talk to the hospital's records." Which specific line in the JSON code proves they are right?
- **(3b) Decoding:** What does the error HIS_Handshake_Timeout tell you about the link between the **Mobile App** and the **Hospital Information System (HIS)**?

Question 3

25 points

Business Majors

- **(3a) The "Why" Factor:** One customer complained about "crashing" or "merchant support." Does Dataset B show any evidence that matches these complaints?
- **(3b) The Full Picture:** Why is it dangerous for TPay to *only* look at the logs and ignore the text feedback?

Healthcare Majors

- **3a) Investigate:** Apex Pharma's Specialty Drugs (Transaction ID: TXN-405) took the longest. Why might "Specialty" items require more time to finalize than standard PPE?
- **(3b) Prioritize:** You only have time to fix one relationship this week. Would you prioritize **Apex Pharma** (Drugs) or **BioLink Lab** (Reagents)? Justify your choice based on clinical risk.

Question 4

25 points

Business Majors

Imagine you are a consultant for TPay Management. Answer the following:

- **(4a) Priority #1:** Based on the data, what is the **one** thing TPay must fix immediately to keep users?

- **(4b) The Future (AI):** How could an AI "Chatbot" or "Sentiment Tool" help TPay handle 10,000 complaints a day?
- **(4c) The "North Star" Metric:** What one number (KPI) should the CEO check every morning? (e.g., Transaction Success Rate, Average Review Score, or Daily Active Users?)

Healthcare Majors

- **(4a) Crisis Management:** You have 50 patients who received incorrect late notices this morning. What is your **first** action to prevent a flood of angry phone calls?
- **(4b) Policy Change:** Propose a "Safety Buffer" policy. For example, how long should the system wait before sending a late notice to allow for potential sync delays?

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Question 1

25 points

Business Majors

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Log Analysis: (Business Majors)

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Log Analysis: (Healthcare Majors)

The Technical Log (Semi-Structured JSON Data)

JSON

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{ "transaction_id": "MOB-8821",
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  "payment_gateway": "Stripe",
  "auth_status": "SUCCESS",
  "processing_details": {
    "gateway_confirmation": "Confirmed",
    "internal_sync_status": "FAILED",
    "error_log": "HIS_Handshake_Timeout"
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- **(3a) Evidence:** Patient B says the system "doesn't talk to the hospital's records." Which specific line in the JSON code proves they are right?
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